

“The storm of shot and shell launched against the Spaniard was destructive beyond all description”

LIEUTENANT J L STICKNEY, EYEWITNESS TO THE BATTLE OF MANILA, 1898

as a fleet of iron-armoured paddle steamers, to fight on the Mississippi. By the conclusion of the Civil War, in 1865, the superiority of ironclad steamships over wooden sailing ships was generally accepted. In 1871, the British built HMS *Devastation*, the first capital warship to have no sails at all. However, a new configuration of naval warfare still took time to emerge: the design of naval guns and their positioning on ships went through a stuttering development. For a time, naval designers became obsessed with rams, imagining that armour would make warships immune to gunfire; they pictured a return to the tactics of Ancient Greece in which steamships manoeuvred like triremes in a bid to sink enemy vessels with their reinforced prows.

Eventually a system was devised for mounting large rifled guns in rotating armoured turrets, with the magazine and ammunition stored in the hull below. These guns fired high-explosive shells at long range, requiring the invention of complex range-finding devices. Hulls were now made from metal and clad in steel armour, but the increasing power of explosives meant that ships remained vulnerable

to this kind of gunfire. In addition, the adoption of steam power was not without its drawbacks. Sailing ships had not needed fuel, but coaling stations were now essential to naval operations, giving a new twist to geopolitics. When coal began to be replaced by oil, the possession of oilfields became a primary strategic concern for naval powers.

NAVAL RACE

By the end of the 19th century, navies had become supreme status symbols for competing imperial powers. The US, Japan, and Germany embarked on large-scale naval construction programmes, to which the UK actively responded, determined to maintain its long-established naval supremacy. In the first decade of the 20th century, a frenzied naval race between the British and the Germans saw the UK repeatedly raise the bar with faster, more powerfully armed battleships, starting with the epoch-making HMS *Dreadnought* in 1906. Yet while naval commanders and a jingoistic public were obsessed with large warships – battleships, battlecruisers, and cruisers – other developments in naval warfare made these expensive vessels worryingly vulnerable. Sea mines, first used in the Crimean War, could sink any capital ship – and so could boats armed with torpedoes, whose worth was ably demonstrated by the Japanese during their war with Russia from 1904 to 1905. Destroyers were devised to defend the fleet against attack by torpedo boats, as well as launching torpedoes themselves, but navies had no technology for defence against torpedo-armed submarines, which became a practical element of navies in the first decade of the 20th century.

KEY BATTLE

THE BATTLE OF MANILA

1 MAY 1898

The Spanish–American War of 1898 was the first test of the American “New Navy” of modern battleships and cruisers. The Asiatic Squadron under Commodore George Dewey sank seven Spanish warships in the Philippines without loss.



▲ American battleships steaming in line destroy Spanish warships in Manila Bay.

▼ WHITEHEAD TORPEDO

In the 1860s, British engineer Robert Whitehead invented the first effective self-propelled torpedo. Carried by small ships, the torpedo threatened to undermine the dominance of heavily gunned warships.



▼ DAHLGREN GUN

Invented by US naval officer John A Dahlgren, the Dahlgren gun was in frequent use during the American Civil War. Its distinctive soda-bottle shape gave more power than a conventional naval gun.

